

1. Array "Equality"



```
[] == ![] //=> true  
//=> Empty array equals NOT empty array!
```

```
[1] == 1 //=> true  
[1,2] == "1,2" //=> true
```

2. The Infinity Paradox



```
Infinity - Infinity //=> NaN  
Infinity / Infinity //=> NaN  
Infinity === Infinity //=> true (at least this makes sense!)  
  
typeof Infinity //=> "number"
```

3. String Multiplication



```
[] == ![] //=> true  
//=> Empty array equals NOT empty array!
```

```
[1] == 1 //=> true  
[1,2] == "1,2" //=> true
```

4. The Void Operator



```
"3" * "3" //=> 9 (multiplies fine)  
"3" + "3" //=> "33" (but addition concatenates)  
"3" - "1" //=> 2 (subtraction works)  
"3" + 1 //=> "31" (but this concatenates again!)
```

5. parseInt Surprises



```
parseInt(0.0000008) //=> 8
// Because it converts to "8e-7" then parses "8"

parseInt(null, 24) //=> 23
//=> "null" in base-24: n=23, u=30, l=21, l=21... takes first valid digit

['1','7','11'].map(parseInt) //=> [1, NaN, 3]
//=> Because map passes (element, index): parseInt('7', 1), parseInt('11', 2)
```

6. Math.max vs Math.min



```
Math.max() //=> -Infinity
Math.min() //=> Infinity
//=> The empty case returns the identity element!

Math.max() < Math.min() //=> true
```

7. The Comma Operator

```
let x = (1, 2, 3)
x //=> 3 (returns the last value)

if (true, false) {
  console.log("won't print")
}
```

8. Tricky Truthy/Falsy

```
Boolean('false') //=> true (non-empty string)
Boolean('') //=> false
Boolean(new Boolean(false)) //=> true (it's an object!)

!!'false' === true //=> true
```

9. Try-Catch-Finally Return



```
function test() {  
  try {  
    return 1  
  } finally {  
    return 2  
  }  
}  
test() //=> 2 (finally overrides the return!)
```



```
function test2() {  
  try {  
    throw "Error"  
  } finally {  
    return "No error!"  
  }  
}  
test2() //=> "No error!" (finally swallows the error)
```

10. Object Key Coercion



```
const obj = {}  
obj[{}] = 1  
obj[{x: 1}] = 2
```

```
obj //=> { "[object Object]": 2 }  
//=> All objects become the same key!
```

```
const key1 = {}  
const key2 = {}  
obj[key1] = 'first'  
obj[key2] = 'second'  
obj[key1] //=> "second" (they're the same key!)
```

Bonus



```
!!null //=> false  
!!"" //=> false  
!!0 //=> false
```

```
//=> But...
```

```
!![] //=> true (empty array is truthy)  
!!{} //=> true (empty object is truthy)
```

```
//=> Yet...
```

```
[] == false //=> true
```